

Mohammad Mashrur Mahtab Mahi

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EDUCATION

iBSc in Computer Science | Lassonde School of Engineering, York University

Toronto, ON

Cumulative GPA 8.27/9.00, Member of Dean's Honour Roll

Graduating Apr 2028

- President's International Scholarship of Excellence & International Award for Academic Excellence (\$190,000 value)
- Leadership: President (BSAY), Founder (Leo Club of Dhaka 400 | Lions International), Operations (SOON Hackathon)

WORK EXPERIENCE

AI Product Development Intern

Apr 2026 – July 2026

Herb Immortal | myherbalshop.ai | herbimmortal.com

Toronto, Canada

- Reduced API response time by **70%** across a MySQL-backed REST API serving **500+** users and **50+** doctors through query optimization, connection pooling, and parameterized requests
- Built and shipped a full-stack AI-powered patient-doctor matching platform live in production, with a Node.js/Express backend, React frontend, JWT + OTP two-factor auth from scratch, and nginx + pm2 deployment on a VPS
- Implemented a structured data pipeline from user input through validation and status tracking to an admin CRM dashboard, covering investor leads, contact submissions, and article management

Research Assistant

Jan 2026 – Apr 2026

York University

Toronto, Canada

- Built Java-based discrete-event simulation modules using OOP to model and analyze blockchain consensus mechanisms, covering event scheduling, queue and state tracking, and system behavior analysis
- Designed hashpower experiments and PoW runtime simulations to evaluate mining behavior patterns, improving local dev workflow using Git, Maven, and JUnit 5

Field Education Program Associate

May 2026 – Present

York University – Faculty of Liberal Arts & Professional Studies

Toronto, Canada

- Identified and quantified **76** platform issues across 6 months, built a formal software replacement proposal with data from **250+** students and **250+** field instructors, and presented findings to school directors and the dean
- Provided technical support with our current software and planning the implementation of InPlace across students

PROJECTS

National Livestock Registry *Node.js | Express | PostgreSQL | Next.js | React Native | TS*

June 2026 – July 2026

- Designed and built a PostgreSQL database with 7 migrations handling real owner PII and animal records at national scale, with parameterized queries and soft deletes to ensure data integrity
- Designed for **2,000-3,000** MAU at national scale, solo-built across web, mobile, and REST API sharing one backend

Delatio - Urban Risk Intelligence Platform *LangGraph | vLLM | llama.cpp | Docker | Python*

May 2026

- Fine-tuned Llama-3.1-Nemotron-70B (FP8 quantized) on 1.2M rows of Open Data Toronto 311 logs and Gemma 4 26B MoE on 45,000 labeled urban infrastructure images, running both simultaneously on a single NVIDIA GB10 node
- Solved concurrent OOM constraints from running a 70B + 26B model stack through aggressive quantization, background weight-paging, and memory-isolated Docker containers with zero cloud dependencies

Student Helper - AI-Powered Learning Platform (1st Place @ BEST Startup Experience)

Sept 2024 – Present

- Built an ML pipeline predicting student final exam scores using Linear Regression, Decision Trees, and Neural Networks trained on real academic performance data from university LMS platforms
- Reached **1,000+** unique logins in 1 summer through direct Knowledge Tracing dashboard built with D3.js and Chart.js

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++

Frameworks: Node.js, Express.js, Next.js, React, React Native, FastAPI, Flask, Tailwind CSS

Databases: PostgreSQL, MySQL, Firebase, Neo4j, MongoDB

AI/ML: TensorFlow, PyTorch, SciKit-Learn, LangGraph, Hugging Face, spaCy, NLTK, Gemini API

Tools: Git, Docker, nginx, pm2, Jest, JUnit 5, Maven, Jira, REST APIs, Vercel

Auth and Security: Auth0, JWT, bcrypt, parameterized queries, httpOnly cookies

Infrastructure: VPS deployment, Render, Neon, Cloudflare, Expo (bare workflow)